Experiment 4: Kronig–Penney Model

Aim

To simulate and study the **Energy Functional vs Energy graph** using the **Kronig–Penney model**.

Software Required

- NanoHUB Online Tool
- Kronig–Penney Model – Periodic Potential Lab

Procedure

1. Open **NanoHUB** online.
2. Go to **Resources → Tools**.
3. Select **Kronig–Penney Model** from the Tags.
4. Open **Kronig–Penney Model → Periodic Potential Lab**.
5. Click **Launch Tool** in the Info section.
6. Select **Potential Type → Step Well**.
7. Under **Energy Details**, set **Maximum Barrier Height = 1 eV**.
8. Click **Simulate** to perform the simulation.
9. Select **Results → Energy Functional vs Energy Theory**.
10. Observe the resulting graph showing the allowed and forbidden energy regions.

Theory

The **Kronig–Penney model**, introduced by Ralph Kronig and William Penney in 1931, is a simplified one-dimensional model used to explain the formation of **energy bands and forbidden energy gaps** in crystalline solids.

The model assumes that an electron moves through a **periodic potential** representing the regularly spaced atoms of a crystal. The potential consists of alternating wells and barriers.

For a periodic potential,

$$ V(x+a)=V(x) $$

where a is the periodicity of the crystal lattice.

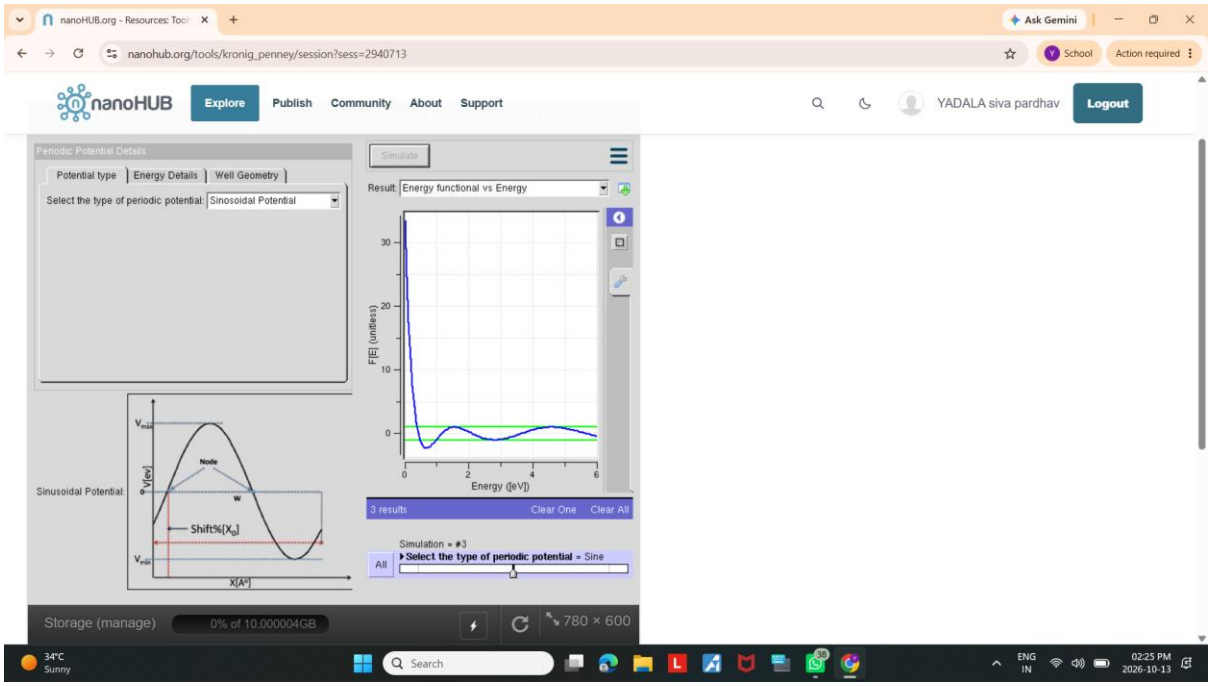
According to **Bloch's theorem**, the electron wave function can be written as

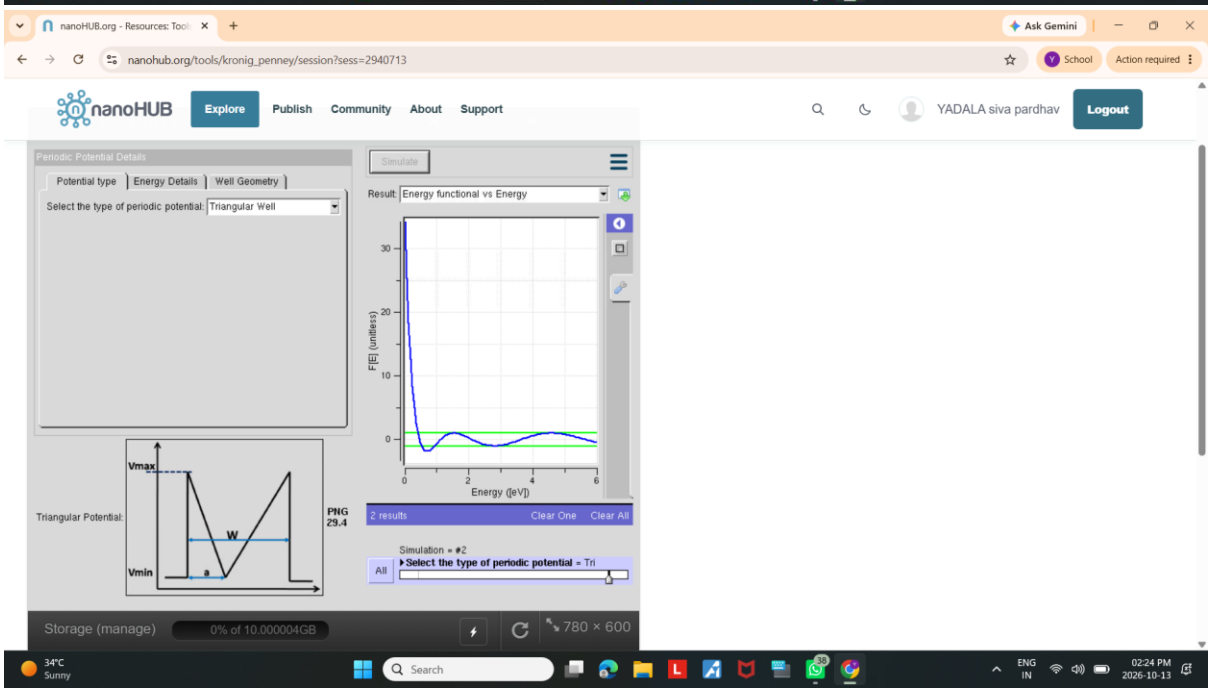
$$ \psi(x)=u(x)e^{ikx} $$

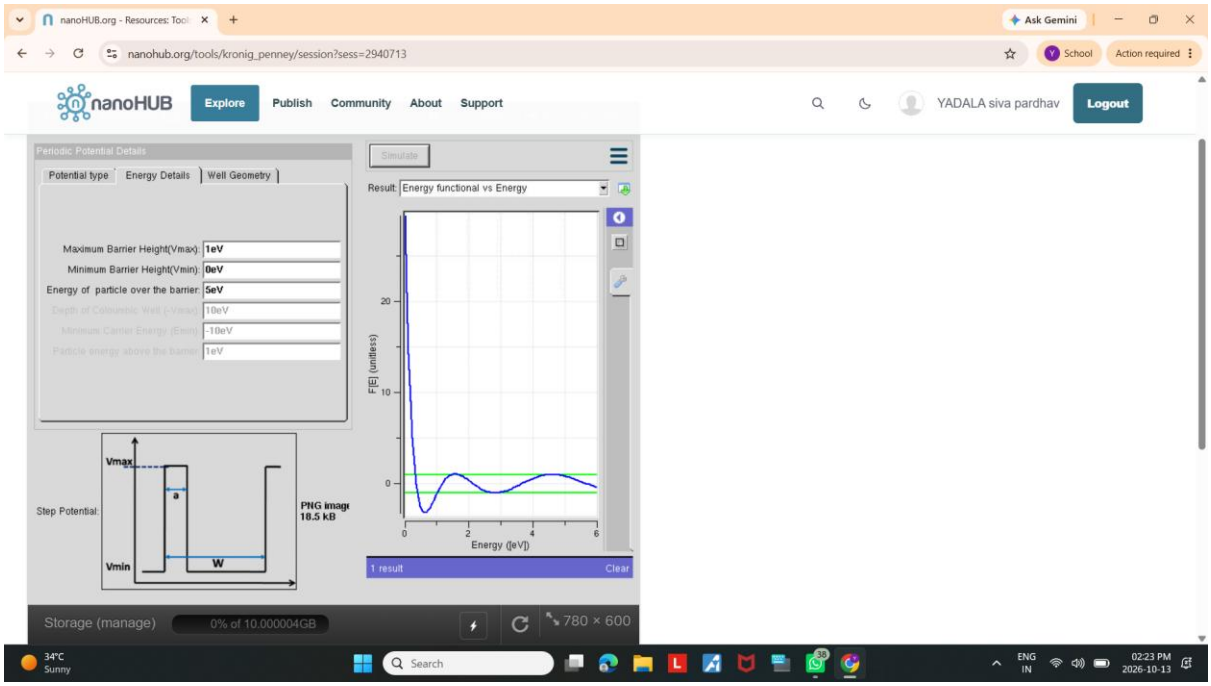
For the Kronig–Penney model, the relationship between energy E and wave vector k determines whether an energy value is **allowed or forbidden**. The regions satisfying the

condition for real \(k\) correspond to allowed energy bands, while the remaining regions correspond to forbidden energy gaps.

Output


nanoHUB.org - Resources: Tools
nanohub.org/tools/kronig_penney/session?sess=2940713
Ask Gemini
School
Action required
nanoHUB
Explore
Publish
Community
About
Support
YADALA siva pardhav
Logout
Periodic Potential Details
Potential type
Energy Details
Well Geometry
Select the type of periodic potential: Sinosoidal Potential
Simulate
Result: Energy functional vs Energy
F[E] (unitless)
Energy ([eV])
Sinusoidal Potential:
Node
Shift%[X0]
X[A°]
3 results
Clear One
Clear All
Simulation = #3
Select the type of periodic potential = Sine
All
Storage (manage)
0% of 10.000004GB
780 × 600
34°C
Sunny
Search
ENG
IN
02:25 PM
2026-10-13


nanoHUB.org - Resources: Tools
nanohub.org/tools/kronig_penney/session?sess=2940713
Ask Gemini
School
Action required
nanoHUB
Explore
Publish
Community
About
Support
YADALA siva pardhav
Logout
Periodic Potential Details
Potential type
Energy Details
Well Geometry
Select the type of periodic potential: Triangular Well
Simulate
Result: Energy functional vs Energy
F[E] (unitless)
Energy ([eV])
Triangular Potential:
Vmax
Vmin
W
a
PNG
29.4
2 results
Clear One
Clear All
Simulation = #2
Select the type of periodic potential = Tri
All
Storage (manage)
0% of 10.000004GB
780 × 600
34°C
Sunny
Search
ENG
IN
02:24 PM
2026-10-13



Expected observation: The graph shows alternating **allowed energy bands** and **forbidden energy gaps**, demonstrating the basic principle of electronic band formation in periodic crystal potentials.

Result

The **Kronig–Penney model was successfully simulated** using the NanoHUB Periodic Potential Lab. The Energy Functional vs Energy result demonstrates the formation of **allowed energy bands and forbidden energy gaps** due to the periodic potential experienced by electrons in a crystal.